Reconnaissance Drought Index Analysis

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```r
knitr::opts_chunk$set(cache = TRUE, cache.lazy = FALSE, warning = FALSE,
                      message = FALSE,echo = TRUE, dpi = 300,
                      fig.width = 15, fig.height = 10)
```

```r
rm(list = ls())
```

```r
library(DT)


library(tidyverse)
library(SPEI)
library(drought)
library(extRemes)
library(heatwaveR)
library(lubridate)
library(fitdistrplus)
```

#Data upload from the folder

```r
data <- readRDS("/Users/mandarphatak/Documents/GitHub/Climate-Analysis/DWD_Folder/Data_DWD/Hamburg_Fuhl
```

Monthly frequency

```r
monthly_tbl <- data |>
  dplyr::mutate(year = year(date),
         month = month(date)) |>
  group_by(year, month) |>
  summarize(
    prec = sum(prec, na.rm = TRUE),
    Tavg = mean(Tavg, na.rm = TRUE),
    Tmin = mean(Tmin, na.rm = TRUE),
    Tmax = mean(Tmax, na.rm = TRUE),
    .groups = "drop"
  ) |>
 dplyr:: mutate(date = make_date(year, month)) |>
  dplyr::select(year, month, date, prec:Tmax)

monthly_tbl <- monthly_tbl |> drop_na()
```

Confirming PET computaiton method:- choice between Hargraves and Thornthwaite

```
## PET computation
PET_hargreaves <- hargreaves(Tmin = monthly_tbl$Tmin,Tmax = monthly_tbl$Tmax,Pre = monthly_tbl$prec,lat
```

```
## [1] "Calculating reference evapotranspiration using the Hargreaves method. Using latitude (`lat`) to
```

```
PET_thornthwaite <- thornthwaite(Tave = monthly_tbl$Tavg,lat = 53.6332)
```

```
## [1] "Checking for missing values (`NA`): all the data must be complete. Input type is vector. Assumi
```

```
# PET_hargreaves |> as_tibble()
# PET_thornthwaite |> as_tibble()

#cbind(PET_hargreaves |> as_tibble() , PET_thornthwaite |> as_tibble())
```

We select Hargreaves

Key Observations from Your Data:

Thornthwaite shows many zero values - This is problematic as it indicates the method fails during winter months when temperatures are low Hargreaves provides continuous values - More realistic for year-round analysis Hargreaves generally shows higher PET values - This is typical for coastal areas

Recommendation for Hamburg Fühlsbuttel: Use Hargreaves method for the following reasons: Climate-Specific Advantages:

Coastal location: Hargreaves performs better in maritime climates like Hamburg Temperature range: Better suited for the moderate temperature variations in northern Germany Humidity considerations: While it doesn't directly use humidity, it's calibrated for regions with moderate humidity

Technical Advantages:

No zero values: Ensures continuous drought index calculation Widely validated: Extensively tested in European climates Robust performance: Less sensitive to extreme temperature variations

For Your Location Specifically:

Hamburg's maritime climate (Köppen Cfb) is well-suited for Hargreaves The method works well for locations with moderate temperature ranges (like northern Germany) Less prone to computational issues in winter

Selecting Hargreaves for the Computation of PET

```
monthly_tbl <- monthly_tbl |>
  mutate(PET = hargreaves(Tmin = monthly_tbl$Tmin,Tmax = monthly_tbl$Tmax,Pre = monthly_tbl$prec,lat = 5
  rename(PREC = prec)
```

```
## [1] "Calculating reference evapotranspiration using the Hargreaves method. Using latitude (`lat`) to
```

Drought Indices:-

Reconnaissance Drought Index (RDI) :- Computation is as follows

```r
# Computation function for RDI
calculate_monthly_rdi <- function(data, scale) {
  require(dplyr)
  require(zoo)

  data <- data %>%
    arrange(date) %>%
    filter(PET > 0, PREC >= 0) %>%
    mutate(
      PREC_sum = rollsum(PREC, k = scale, fill = NA, align = "right"),
      PET_sum = rollsum(PET, k = scale, fill = NA, align = "right"),
      alpha = ifelse(PET_sum > 0, PREC_sum / PET_sum, NA_real_)
    )

  valid_alphas <- data$alpha[!is.na(data$alpha) & is.finite(data$alpha) & data$alpha > 0]

  if (length(valid_alphas) > 30) {
    log_alphas <- log(valid_alphas)
    y_bar <- mean(log_alphas, na.rm = TRUE)
    sigma_y <- sd(log_alphas, na.rm = TRUE)

    data <- data %>%
      mutate(
        RDIst = ifelse(alpha > 0, (log(alpha) - y_bar) / sigma_y, NA_real_)
      )
  } else {
    data$RDIst <- NA_real_
  }

  # Only return date and RDIst
  return(data |>  dplyr::select(date, RDIst))
}
# Funciton for differrent scales
calculate_rdist_for_scales <- function(data, scales = c(3,6,9,12,24,36,48)) {
  require(dplyr)

  result <- data |> dplyr:: select(date)  # Start with date column

  for (k in scales) {
    rdi_k <- calculate_monthly_rdi(data, scale = k) %>%
      rename(!!paste0("RDI_", k) := RDIst)

    result <- left_join(result, rdi_k, by = "date")
  }

  return(result)
}
```

```r
monthly_rdist_tbl <- calculate_rdist_for_scales(data = monthly_tbl)
```

We have 3,6,9,12,24,36,48 monthly scale. So then which scales are important for the analysis. Since the period is 1936-2025, so 3,6,9 scales will not be that informative as these will refelect very short term changes. So 12,24,36,48 are better choices

RDI 12

```r
library(dplyr)

rdi_12tbl <- monthly_rdist_tbl |>
  dplyr::select(date,RDI_12) |>
  drop_na()
# Recent period chosen 1990-2025
rdi_12tbl <- rdi_12tbl |>
  dplyr::mutate(sign = ifelse(RDI_12 >0,"pos","neg")) |>
  filter(date>"1989-12-01") |>
# Enhanced data preparation with drought categories
 dplyr:: mutate(
    # Original sign classification
    sign = ifelse(RDI_12 > 0, "pos", "neg"),

    drought_category = case_when(
      RDI_12 >= 2.0 ~ "Extremely Wet",
      RDI_12 >= 1.5 & RDI_12 < 2.0 ~ "Very Wet",
      RDI_12 >= 1.0 & RDI_12 < 1.5 ~ "Moderately Wet",
      RDI_12 >= 0.5 & RDI_12 < 1.0 ~ "Slightly Wet",
      RDI_12 >= -0.5 & RDI_12 < 0.5 ~ "Normal",
      RDI_12 >= -1.0 & RDI_12 < -0.5 ~ "Mild Drought",
      RDI_12 >= -1.5 & RDI_12 < -1.0 ~ "Moderate Drought",
      RDI_12 >= -2.0 & RDI_12 < -1.5 ~ "Severe Drought",
      TRUE ~ "Extreme Drought"  # RDI_12 < -2.0
    ),

    # Create ordered factor for proper legend ordering
    drought_category = factor(drought_category, levels = c(
      "Extreme Drought", "Severe Drought", "Moderate Drought",
      "Mild Drought", "Normal", "Slightly Wet",
      "Moderately Wet", "Very Wet", "Extremely Wet"
    ))
  )

# Calculate key statistics for plot annotations
drought_stats <- rdi_12tbl |>
  summarise(
    total_months = n(),
    drought_months = sum(RDI_12 < -0.5),
```

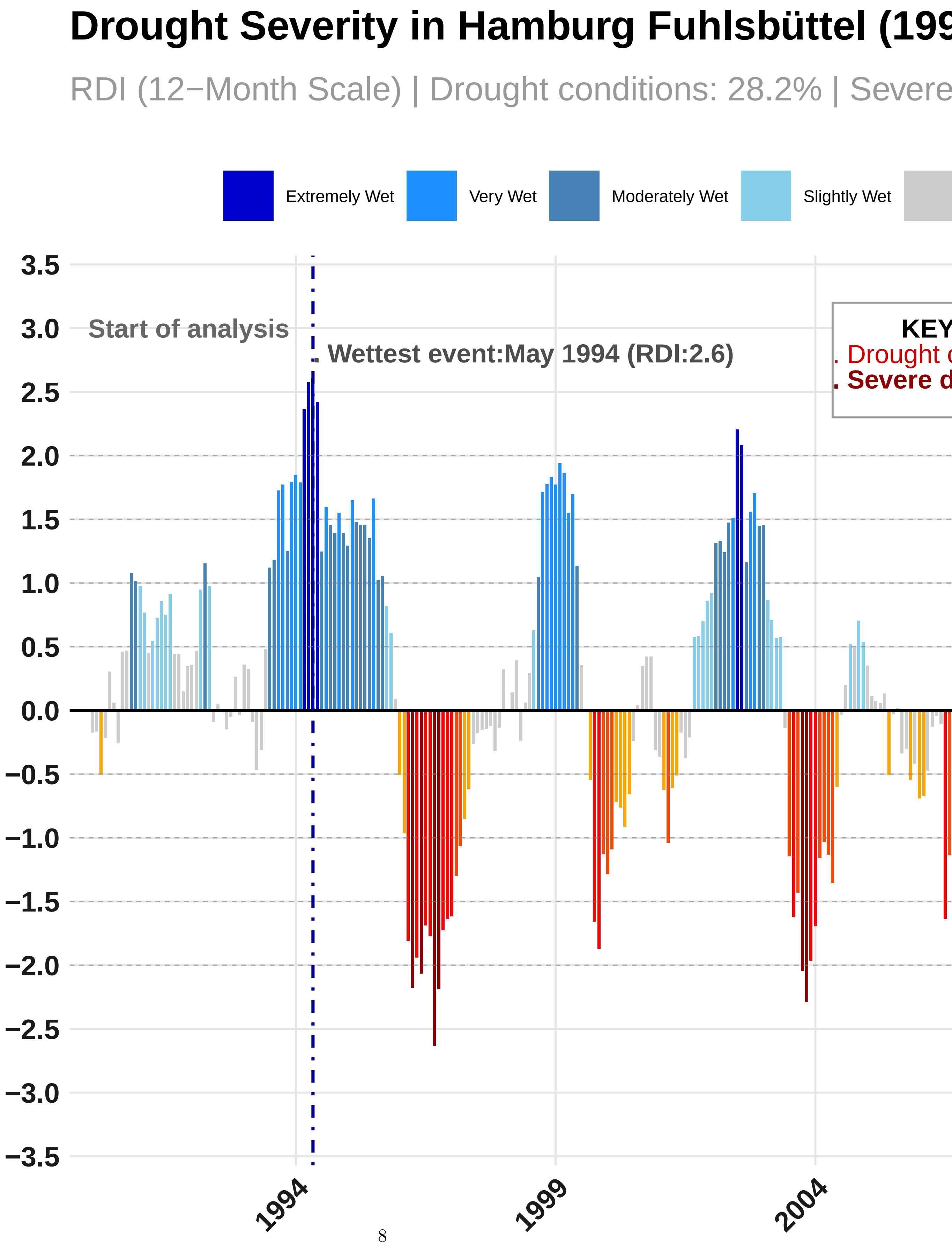
```r
    severe_drought_months = sum(RDI_12 < -1.5),
    wet_months = sum(RDI_12 > 0.5),
    min_rdi = min(RDI_12),
    max_rdi = max(RDI_12),
    min_date = date[which.min(RDI_12)],
    max_date = date[which.max(RDI_12)],
    drought_pct = round(drought_months / total_months * 100, 1),
    severe_drought_pct = round(severe_drought_months / total_months * 100, 1),
    wet_pct = round(wet_months / total_months * 100, 1)
  )

# Create statistical summary for subtitle
stats_text <- paste0(
  "Drought conditions: ", drought_stats$drought_pct, "% | ",
  "Severe drought: ", drought_stats$severe_drought_pct, "% | ",
  "Wettest: ", format(drought_stats$max_date, "%b %Y"), " (", round(drought_stats$max_rdi, 1), ") | ",
  "Driest: ", format(drought_stats$min_date, "%b %Y"), " (", round(drought_stats$min_rdi, 1), ")"
)

# Define color palette for drought categories
drought_colors <- c(
  "Extreme Drought" = "#8B0000",     # Dark Red
  "Severe Drought" = "#FF0000",      # Red
  "Moderate Drought" = "#FF4500",    # Orange Red
  "Mild Drought" = "#FFA500",        # Orange
  "Normal" = "#CCCCCC",              # Light Gray
  "Slightly Wet" = "#87CEEB",        # Sky Blue
  "Moderately Wet" = "#4682B4",      # Steel Blue
  "Very Wet" = "#1E90FF",            # Dodger Blue
  "Extremely Wet" = "#0000CD"        # Medium Blue
)
```

Visualization of RDI 12

```r
p0 <- ggplot(rdi_12tbl, aes(x = date, y = RDI_12, fill = drought_category)) +
  geom_col(show.legend = TRUE, width = 20) +
  geom_hline(yintercept = 0, color = "black", linewidth = 0.8) +

  # Add reference dashed lines
  geom_hline(yintercept = c(-2, -1.5, -1, -0.5, 0.5, 1, 1.5, 2),
             color = "grey50", linewidth = 0.3, linetype = "dashed", alpha = 0.6) +

  # Highlight driest and wettest periods
  geom_vline(xintercept = drought_stats$min_date, color = "#8B0000", linetype = "dotdash", linewidth = 
  # geom_text(aes(x = drought_stats$min_date, y = drought_stats$min_rdi - 0.3),
  #           label = paste0("Driest: ", format(drought_stats$min_date, "%b %Y")),
  #           angle = 90, hjust = 1.1, size = 3.3, color = "#8B0000") +

  geom_vline(xintercept = drought_stats$max_date, color = "#00008B", linetype = "dotdash", linewidth = 
  # geom_text(aes(x = drought_stats$max_date, y = drought_stats$max_rdi + 0.3),
  #           label = paste0("Wettest: ", format(drought_stats$max_date, "%b %Y")),
  #           angle = 90, hjust = -0.1, size = 3.3, color = "#00008B") +
```

```
# KEY INSIGHTS annotation box
  annotate("rect",
           xmin = as.Date("2004-05-01"), xmax = as.Date("2013-03-01"),
           ymin = 2.3, ymax = 3.2,
           fill = "white", color = "grey60", alpha = 0.9, linewidth = 0.5) +

  annotate("text", x = as.Date("2005-01-01"), y = 3.0,
           label = paste0("KEY INSIGHTS (", min(year(rdi_12tbl$date)), "-", max(year(rdi_12tbl$date)),
           size = 5, fontface = "bold", hjust = -0.1,) +

  annotate("text", x = as.Date("2004-05-01"), y = 2.8,
           label = paste0("• Drought conditions: ", drought_stats$drought_pct, "% of period"),
           size = 5, hjust = 0, color = "#CC0000") +

  annotate("text", x = as.Date("2004-05-01"), y = 2.6,
           label = paste0("• Severe drought: ", drought_stats$severe_drought_pct, "% of period"),
           size = 5, hjust = 0, color = "#8B0000",fontface="bold") +

  annotate("text", x = as.Date("2019-01-01"), y = -3,
           label = paste0(" Event Extreme : ", format(drought_stats$min_date, "%b %Y"),
                         " (RDI:",round(drought_stats$min_rdi, 1), ") •"),
           size = 5, hjust = 1, color = "grey30",fontface="bold") +
annotate("text", x = drought_stats$max_date, y = drought_stats$max_rdi + 0.2,
         label = paste0("• Wettest event:",format(drought_stats$max_date, "%b %Y"),
                       " (RDI:",round(drought_stats$max_rdi, 1),")"),
         size = 5, hjust = 0, color = "grey30", fontface="bold") +
  # Start and end of analysis period
  annotate("text", x = min(rdi_12tbl$date), y = 3,
           label = "Start of analysis", hjust = 0, size = 5, color = "grey40",fontface="bold") +
  annotate("text", x = max(rdi_12tbl$date), y = 3,
           label = "End of analysis", hjust = .9, size = 5, color = "grey50",fontface="bold") +

  # Scales
  scale_x_date(
    date_breaks = "5 years",
    date_labels = "%Y",
    limits = as.Date(c("1990-01-01", "2025-05-28")),
    expand = expansion(mult = c(0.01, 0.01))
  ) +

  scale_y_continuous(
    limits = c(-3.5,3.5),
    breaks = seq(-3.5, 3.5, 0.5),
    expand = expansion(mult = c(0.01, 0.01))
  ) +

  scale_fill_manual(
    name = "Condition",
    values = drought_colors,
    guide = guide_legend(
      title.position = "top",
      title.hjust = 0.5,
      nrow = 1,
```

```r
      reverse = TRUE
    )
  ) +

  # Labels and theme
  labs(
    title = "Drought Severity in Hamburg Fuhlsbüttel (1990-2025*)",
    subtitle = paste0("RDI (12-Month Scale) | ", stats_text),
    caption = "Source: DWD ",
    x = "",
    y = ""
  ) +
  theme_minimal(base_size = 12) +
  theme(

    panel.grid.minor.x = element_blank(),
    panel.grid.major.x = element_line(color = "grey90", linewidth = 0.5),
    panel.grid.minor.y = element_blank(),
    panel.grid.major.y = element_line(color = "grey90", linewidth = 0.5),
    plot.title = element_text(face = "bold", size = 22, margin = margin(t = 10, b = 15)),
    plot.subtitle = element_text(color = "grey60", size = 18, margin = margin(b = 10)),
    plot.caption = element_text(color = "grey60", size = 15, hjust = 1, margin = margin(t=10)),
    axis.text.x = element_text(angle = 45, hjust = 1, size = 15, colour = "grey10",face = "bold"),
    axis.text.y = element_text(size = 15, colour = "gray10",face="bold"),
    axis.title.y = element_text(size = 16, margin = margin(r = 10)),
    legend.position = "top",
    legend.title = element_text(size = 12, face = "bold"),
    legend.text = element_text(size = 9),
    legend.key.size = unit(1, "cm"),
    #plot.margin = margin(t = 20, r = 20, b = 20, l = 20)
  )
print(p0)
```



Drought Severity in Hamburg Fuhlsbüttel (199
RDI (12−Month Scale) | Drought conditions: 28.2% | Severe
Extremely Wet
Very Wet
Moderately Wet
Slightly Wet
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
−3.0
−3.5
Start of analysis
. Wettest event:May 1994 (RDI:2.6)
KEY
. Drought c
. Severe d
1994
1999
2004

```r
# Saving the plot
 ggsave("/Users/mandarphatak/Documents/GitHub/Climate-Analysis/DWD_Folder/Plots/RDI_12_Hamburg.png", pl
```

Inference and Analysis: The 12-month RDI analysis for Hamburg Fuhlsbüttel from 1990 to 2025 reveals a shifting pattern in annual drought dynamics. While the early 1990s experienced substantial wet conditions—including the wettest period in May 1994 (RDI = 2.6)—recent decades have seen more frequent and intense droughts, culminating in the extreme event of January 2019 (RDI = -3.0). Overall, 28.2% of the period was marked by drought (RDI < -0.5), and 7.8% experienced severe drought (RDI < -1.5), indicating a significant increase in water stress risk, particularly in the last decade. These findings emphasize the importance of ongoing monitoring and adaptation strategies, especially in light of potential climate-driven shifts in hydrological cycles.

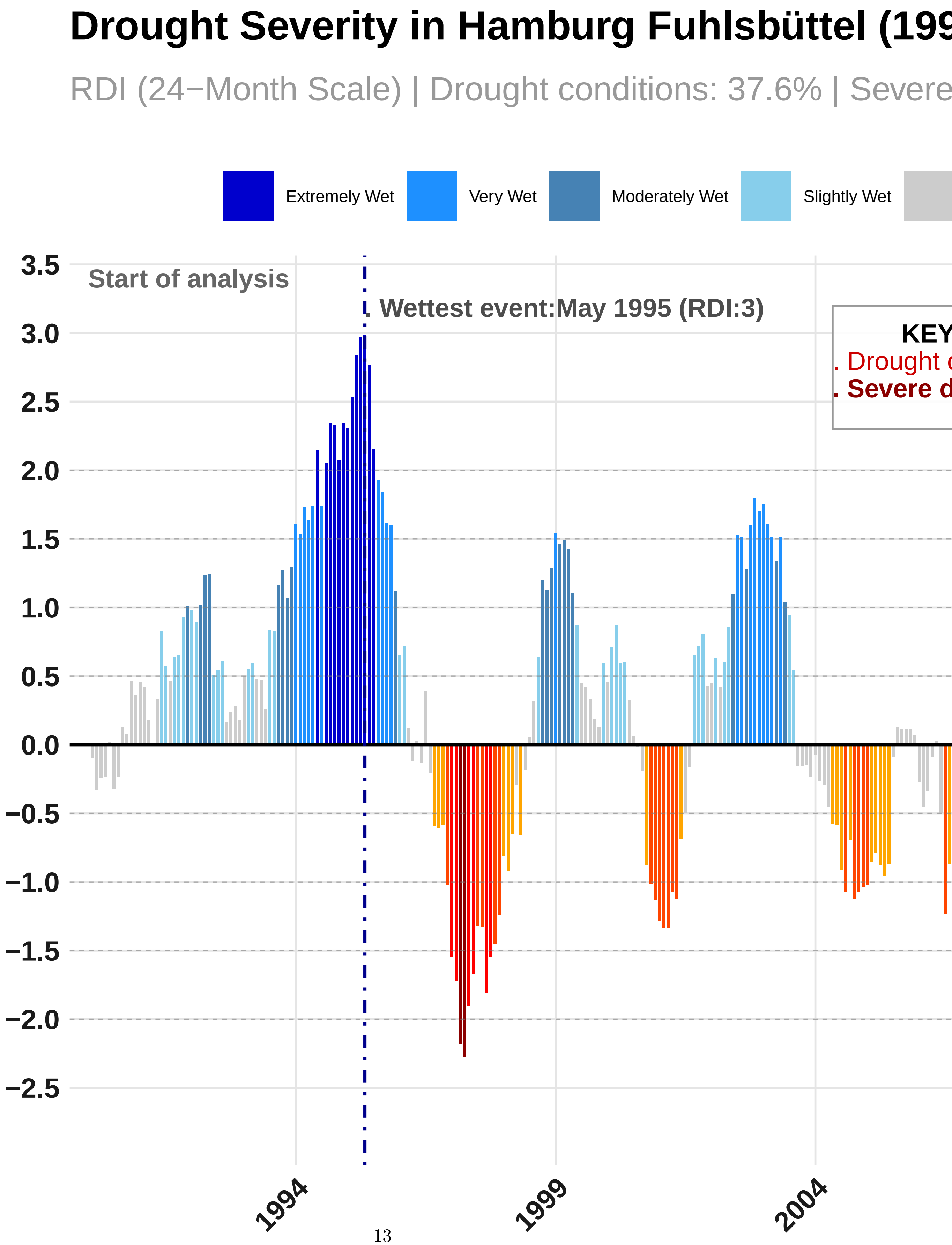
RDI 24

```r
#
# Enhanced data preparation with drought categories
rdi_24tbl <- monthly_rdist_tbl |>
  dplyr::select(date, RDI_24) |>
  drop_na() |>
  filter(date > "1989-12-01") |>
 dplyr:: mutate(
    # Original sign classification
    sign = ifelse(RDI_24 > 0, "pos", "neg"),

    drought_category = case_when(
      RDI_24 >= 2.0 ~ "Extremely Wet",
      RDI_24 >= 1.5 & RDI_24 < 2.0 ~ "Very Wet",
      RDI_24 >= 1.0 & RDI_24 < 1.5 ~ "Moderately Wet",
      RDI_24 >= 0.5 & RDI_24 < 1.0 ~ "Slightly Wet",
      RDI_24 >= -0.5 & RDI_24 < 0.5 ~ "Normal",
      RDI_24 >= -1.0 & RDI_24 < -0.5 ~ "Mild Drought",
      RDI_24 >= -1.5 & RDI_24 < -1.0 ~ "Moderate Drought",
      RDI_24 >= -2.0 & RDI_24 < -1.5 ~ "Severe Drought",
      TRUE ~ "Extreme Drought"  # RDI_24 < -2.0
    ),

    # Create ordered factor for proper legend ordering
    drought_category = factor(drought_category, levels = c(
      "Extreme Drought", "Severe Drought", "Moderate Drought",
      "Mild Drought", "Normal", "Slightly Wet",
      "Moderately Wet", "Very Wet", "Extremely Wet"
    ))
  )

# Calculate key statistics for plot annotations
drought_stats <- rdi_24tbl |>
  summarise(
    total_months = n(),
    drought_months = sum(RDI_24 < -0.5),
    severe_drought_months = sum(RDI_24 < -1.5),
    wet_months = sum(RDI_24 > 0.5),
```

```r
    min_rdi = min(RDI_24),
    max_rdi = max(RDI_24),
    min_date = date[which.min(RDI_24)],
    max_date = date[which.max(RDI_24)],
    drought_pct = round(drought_months / total_months * 100, 1),
    severe_drought_pct = round(severe_drought_months / total_months * 100, 1),
    wet_pct = round(wet_months / total_months * 100, 1)
  )

# Create statistical summary for subtitle
stats_text <- paste0(
  "Drought conditions: ", drought_stats$drought_pct, "% | ",
  "Severe drought: ", drought_stats$severe_drought_pct, "% | ",
  "Wettest: ", format(drought_stats$max_date, "%b %Y"), " (", round(drought_stats$max_rdi, 1), ") | ",
  "Driest: ", format(drought_stats$min_date, "%b %Y"), " (", round(drought_stats$min_rdi, 1), ")"
)

# Define color palette for drought categories
drought_colors <- c(
  "Extreme Drought" = "#8B0000",     # Dark Red
  "Severe Drought" = "#FF0000",      # Red
  "Moderate Drought" = "#FF4500",    # Orange Red
  "Mild Drought" = "#FFA500",        # Orange
  "Normal" = "#CCCCCC",              # Light Gray
  "Slightly Wet" = "#87CEEB",        # Sky Blue
  "Moderately Wet" = "#4682B4",      # Steel Blue
  "Very Wet" = "#1E90FF",            # Dodger Blue
  "Extremely Wet" = "#0000CD"        # Medium Blue
)


# Plot
p1 <- ggplot(rdi_24tbl, aes(x = date, y = RDI_24, fill = drought_category)) +
  geom_col(show.legend = TRUE, width = 20) +
  geom_hline(yintercept = 0, color = "black", linewidth = 0.8) +

  # Add reference dashed lines
  geom_hline(yintercept = c(-2, -1.5, -1, -0.5, 0.5, 1, 1.5, 2),
             color = "grey50", linewidth = 0.3, linetype = "dashed", alpha = 0.6) +

  # Highlight driest and wettest periods
  geom_vline(xintercept = drought_stats$min_date, color = "#8B0000", linetype = "dotdash", linewidth = 
  # geom_text(aes(x = drought_stats$min_date, y = drought_stats$min_rdi - 0.3),
  #           label = paste0("Driest: ", format(drought_stats$min_date, "%b %Y")),
  #           angle = 90, hjust = 1.1, size = 3.3, color = "#8B0000") +

  geom_vline(xintercept = drought_stats$max_date, color = "#00008B", linetype = "dotdash", linewidth = 
  # geom_text(aes(x = drought_stats$max_date, y = drought_stats$max_rdi + 0.3),
  #           label = paste0("Wettest: ", format(drought_stats$max_date, "%b %Y")),
  #           angle = 90, hjust = -0.1, size = 3.3, color = "#00008B") +

  # KEY INSIGHTS annotation box
  annotate("rect",
```

```r
           xmin = as.Date("2004-05-01"), xmax = as.Date("2013-03-01"),
           ymin = 2.3, ymax = 3.2,
           fill = "white", color = "grey60", alpha = 0.9, linewidth = 0.5) +

  annotate("text", x = as.Date("2005-01-01"), y = 3.0,
           label = paste0("KEY INSIGHTS (", min(year(rdi_24tbl$date)), "-", max(year(rdi_24tbl$date)),
           size = 5, fontface = "bold", hjust = -0.1,) +

  annotate("text", x = as.Date("2004-05-01"), y = 2.8,
           label = paste0("• Drought conditions: ", drought_stats$drought_pct, "% of period"),
           size = 5, hjust = 0, color = "#CC0000") +

  annotate("text", x = as.Date("2004-05-01"), y = 2.6,
           label = paste0("• Severe drought: ", drought_stats$severe_drought_pct, "% of period"),
           size = 5, hjust = 0, color = "#8B0000",fontface="bold") +

  annotate("text", x = as.Date("2020-01-01"), y = -2.4,
           label = paste0(" Event Extreme : ", format(drought_stats$min_date, "%b %Y"),
                          " (RDI:",round(drought_stats$min_rdi, 1), ") •"),
           size = 5, hjust = 1, color = "grey30",fontface="bold") +
annotate("text", x = drought_stats$max_date, y = drought_stats$max_rdi + 0.2,
         label = paste0("• Wettest event:",format(drought_stats$max_date, "%b %Y"),
                        " (RDI:",round(drought_stats$max_rdi, 1),")"),
         size = 5, hjust = 0, color = "grey30", fontface="bold") +
  # Start and end of analysis period
  annotate("text", x = min(rdi_24tbl$date), y = 3.4,
           label = "Start of analysis", hjust = 0, size = 5, color = "grey40",fontface="bold") +
  annotate("text", x = max(rdi_24tbl$date), y = 3.4,
           label = "End of analysis", hjust = .9, size = 5, color = "grey50",fontface="bold") +

  # Scales
  scale_x_date(
    date_breaks = "5 years",
    date_labels = "%Y",
    limits = as.Date(c("1990-01-01", "2025-05-28")),
    expand = expansion(mult = c(0.01, 0.01))
  ) +

  scale_y_continuous(
    limits = c(-3,3.5),
    breaks = seq(-2.5, 3.5, 0.5),
    expand = expansion(mult = c(0.01, 0.01))
  ) +

  scale_fill_manual(
    name = "Condition",
    values = drought_colors,
    guide = guide_legend(
      title.position = "top",
      title.hjust = 0.5,
      nrow = 1,
      reverse = TRUE
    )
```

```r
  ) +

  # Labels and theme
  labs(
    title = "Drought Severity in Hamburg Fuhlsbüttel (1990-2025*)",
     subtitle = paste0("RDI (24-Month Scale) | ", stats_text),
    caption = "Source: DWD ",
    x = "",
    y = ""
  ) +

  theme_minimal(base_size = 12) +
  theme(

    panel.grid.minor.x = element_blank(),
    panel.grid.major.x = element_line(color = "grey90", linewidth = 0.5),
    panel.grid.minor.y = element_blank(),
    panel.grid.major.y = element_line(color = "grey90", linewidth = 0.5),
    plot.title = element_text(face = "bold", size = 22, margin = margin(t = 10, b = 15)),
    plot.subtitle = element_text(color = "grey60", size = 18, margin = margin(b = 10)),
    plot.caption = element_text(color = "grey60", size = 15, hjust = 1, margin = margin(t=10)),
    axis.text.x = element_text(angle = 45, hjust = 1, size = 15, colour = "grey10",face = "bold"),
    axis.text.y = element_text(size = 15, colour = "gray10",face="bold"),
    axis.title.y = element_text(size = 16, margin = margin(r = 10)),
    legend.position = "top",
    legend.title = element_text(size = 12, face = "bold"),
    legend.text = element_text(size = 9),
    legend.key.size = unit(1, "cm"),
    #plot.margin = margin(t = 10, r = 20, b = 10, l = 20)
  )

# Print the final plot
print(p1)
```


Drought Severity in Hamburg Fuhlsbüttel (199
RDI (24−Month Scale) | Drought conditions: 37.6% | Severe
Extremely Wet
Very Wet
Moderately Wet
Slightly Wet
Start of analysis
Wettest event:May 1995 (RDI:3)
KEY
Drought
Severe d
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
1994
1999
2004

```r
ggsave("/Users/mandarphatak/Documents/GitHub/Climate-Analysis/DWD_Folder/Plots/RDI_24_Hamburg.png", plo


# # Create detailed summary table
# drought_summary <- rdi_24tbl |>
#   count(drought_category, name = "months") |>
#   mutate(
#     percentage = round(months / sum(months) * 100, 1),
#     total_months = sum(months)
#   ) |>
#   arrange(match(drought_category, levels(rdi_24tbl$drought_category)))
#
# # Print summary statistics
# cat("\n=== DROUGHT ANALYSIS SUMMARY ===\n")
# cat("Period:", min(rdi_24tbl$date), "to", max(rdi_24tbl$date), "\n")
# cat("Total months analyzed:", nrow(rdi_24tbl), "\n\n")
#
# cat("EXTREME EVENTS:\n")
# cat("• Driest period:", format(drought_stats$min_date, "%B %Y"),
#     "with RDI =", round(drought_stats$min_rdi, 2), "\n")
# cat("• Wettest period:", format(drought_stats$max_date, "%B %Y"),
#     "with RDI =", round(drought_stats$max_rdi, 2), "\n\n")
#
# cat("CONDITION DISTRIBUTION:\n")
# print(drought_summary)
#
# cat("\nKEY INSIGHTS:\n")
# cat("• Drought conditions (RDI < -0.5):", drought_stats$drought_pct, "% of the period\n")
# cat("• Severe drought (RDI < -1.5):", drought_stats$severe_drought_pct, "% of the period\n")
# cat("• Wet conditions (RDI > 0.5):", drought_stats$wet_pct, "% of the period\n")
```

Interpretation & Analysis

The 24-month RDI trend for Hamburg Fuhlsbüttel from 1990 to 2024 reveals significant interannual variability in drought and wet conditions. While early decades (especially the 1990s) showed frequent and intense wet periods, the most recent decade has been characterized by prolonged dry conditions, culminating in the driest month observed in January 2020 (RDI = -2.4). Overall, drought conditions were present during 37.6% of the study period, with 4.7% classified as severe. This indicates increasing susceptibility to long-term water stress, especially post-2010.

Comparison of RDI 12 & RDI 24

Metric RDI 12-Month RDI 24-Month Interpretation Period 1990–2025* 1990–2025* Same base period Drought Conditions (< -0.5) 28.2% 37.6% 24-month shows more persistent dryness Severe Drought (< -1.5) 7.8% 4.7% 12-month captures sharper, shorter extremes Wet Conditions (> 0.5) 27.1% 28.9% Fairly similar in wet distribution Driest Event Jan 2019 (RDI = -3.0) Jan 2020 (RDI = -2.4) 12-month shows sharper negative anomaly Wettest Event May 1994 (RDI = 2.6) May 1995 (RDI = 3.0) Similar, offset by 1 year

Interpretation Summary: While the 24-month RDI highlights more persistent drought conditions across multi-year scales, the 12-month RDI is more sensitive to year-to-year variability, capturing sharper extremes like the 2019 drought. The higher frequency of severe drought in the 12-month scale (7.8% vs 4.7%) suggests greater short-term hydroclimatic variability, even though total wet conditions are fairly consistent across both indices.

Part 2: Trend Analysis – Mann-Kendall Test

```r
library(Kendall)

# Ensure the data is in time order and remove NAs
rdi_12_values <- rdi_12tbl |> arrange(date) |> pull(RDI_12) |> na.omit()

# Run Mann-Kendall test
mk_test_12 <- MannKendall(rdi_12_values)
print(mk_test_12)
```

```
## tau = -0.163, 2-sided pvalue =5.4873e-07
```

```r
#Do the same for RDI 24-month:


rdi_24_values <- rdi_24tbl |> arrange(date) |> pull(RDI_24) |> na.omit()
mk_test_24 <- MannKendall(rdi_24_values)
print(mk_test_24)
```

```
## tau = -0.282, 2-sided pvalue =< 2.22e-16
```

The Mann-Kendall trend analysis reveals significant negative trends in both the 12-month and 24-month RDI series for Hamburg Fuhlsbüttel (1990–2025). The 12-month RDI exhibits a moderate downward trend (= -0.163, $p < 0.001$), indicating increasing frequency of dry years. More notably, the 24-month RDI shows a strong and highly significant downward trend (= -0.282, $p < 0.0001$), suggesting an intensification of long-term drought conditions. These results support the evidence of a drying trend in the region, particularly after 2000.

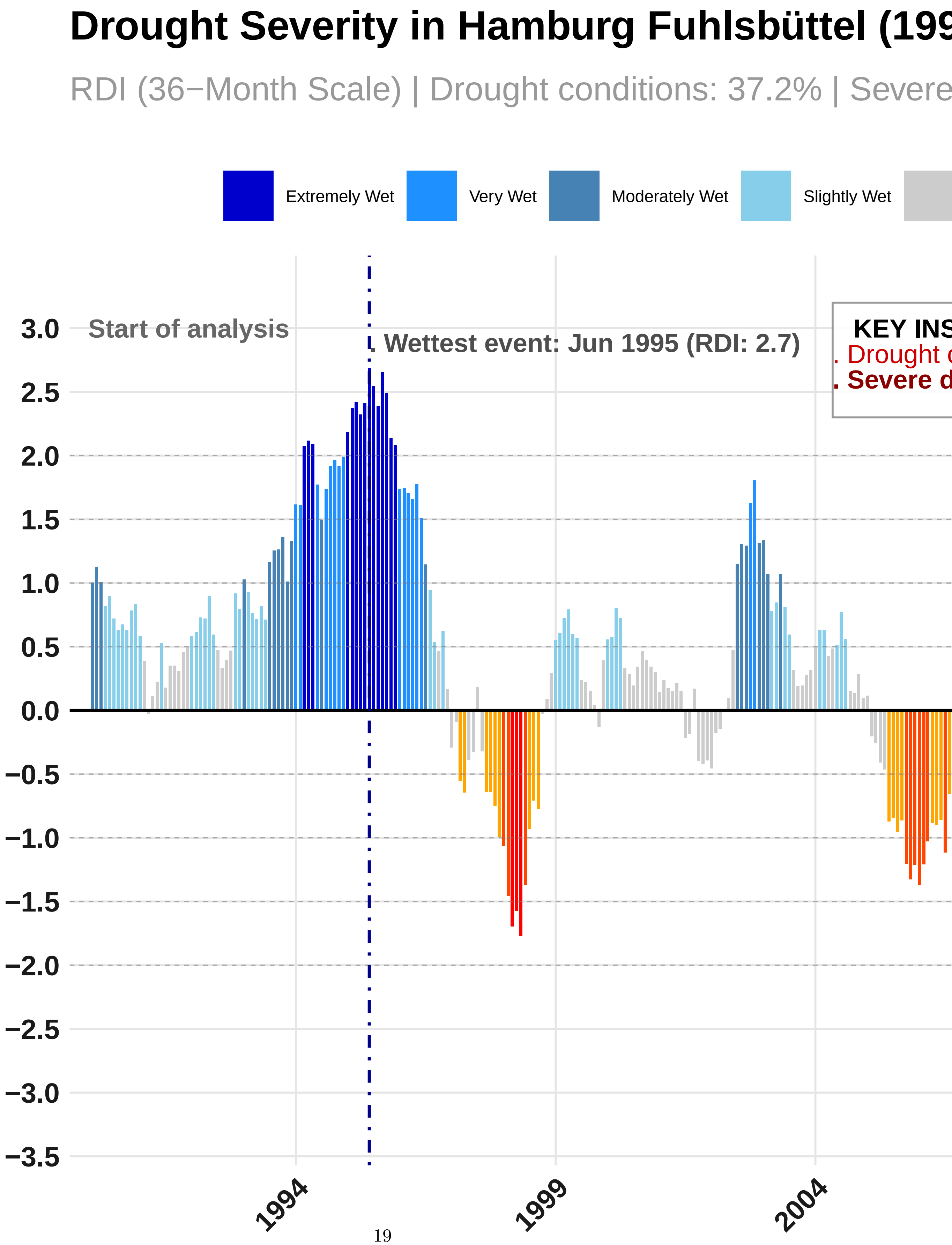
RDI 36

```r
# Data preparation
rdi_36tbl <- monthly_rdist_tbl |>
  dplyr::select(date, RDI_36) |>
  drop_na() |>
  filter(date > "1989-12-01") |>
  dplyr::mutate(
    # Original sign classification
    sign = ifelse(RDI_36 > 0, "pos", "neg"),

    drought_category = case_when(
      RDI_36 >= 2.0 ~ "Extremely Wet",
      RDI_36 >= 1.5 & RDI_36 < 2.0 ~ "Very Wet",
      RDI_36 >= 1.0 & RDI_36 < 1.5 ~ "Moderately Wet",
      RDI_36 >= 0.5 & RDI_36 < 1.0 ~ "Slightly Wet",
      RDI_36 >= -0.5 & RDI_36 < 0.5 ~ "Normal",
      RDI_36 >= -1.0 & RDI_36 < -0.5 ~ "Mild Drought",
      RDI_36 >= -1.5 & RDI_36 < -1.0 ~ "Moderate Drought",
      RDI_36 >= -2.0 & RDI_36 < -1.5 ~ "Severe Drought",
      TRUE ~ "Extreme Drought"  # RDI_36 < -2.0
```

```r
    ),

    # Create ordered factor for proper legend ordering
    drought_category = factor(drought_category, levels = c(
      "Extreme Drought", "Severe Drought", "Moderate Drought",
      "Mild Drought", "Normal", "Slightly Wet",
      "Moderately Wet", "Very Wet", "Extremely Wet"
    ))
  )

# Calculate key statistics for plot annotations
drought_stats <- rdi_36tbl |>
  summarise(
    total_months = n(),
    drought_months = sum(RDI_36 < -0.5),
    severe_drought_months = sum(RDI_36 < -1.5),
    wet_months = sum(RDI_36 > 0.5),
    min_rdi = min(RDI_36),
    max_rdi = max(RDI_36),
    min_date = date[which.min(RDI_36)],
    max_date = date[which.max(RDI_36)],
    drought_pct = round(drought_months / total_months * 100, 1),
    severe_drought_pct = round(severe_drought_months / total_months * 100, 1),
    wet_pct = round(wet_months / total_months * 100, 1)
  )

# Create statistical summary for subtitle
stats_text <- paste0(
  "Drought conditions: ", drought_stats$drought_pct, "% | ",
  "Severe drought: ", drought_stats$severe_drought_pct, "% | ",
  "Wettest: ", format(drought_stats$max_date, "%b %Y"), " (", round(drought_stats$max_rdi, 1), ") | ",
  "Driest: ", format(drought_stats$min_date, "%b %Y"), " (", round(drought_stats$min_rdi, 1), ")"
)

# Define color palette for drought categories
drought_colors <- c(
  "Extreme Drought" = "#8B0000",      # Dark Red
  "Severe Drought" = "#FF0000",       # Red
  "Moderate Drought" = "#FF4500",     # Orange Red
  "Mild Drought" = "#FFA500",         # Orange
  "Normal" = "#CCCCCC",               # Light Gray
  "Slightly Wet" = "#87CEEB",         # Sky Blue
  "Moderately Wet" = "#4682B4",       # Steel Blue
  "Very Wet" = "#1E90FF",             # Dodger Blue
  "Extremely Wet" = "#0000CD"         # Medium Blue
)

# Create the plot
ggplot(rdi_36tbl, aes(x = date, y = RDI_36, fill = drought_category)) +
  geom_col(show.legend = TRUE, width = 20) +
  geom_hline(yintercept = 0, color = "black", linewidth = 0.8) +

  # Add reference dashed lines
```

```r
geom_hline(yintercept = c(-2, -1.5, -1, -0.5, 0.5, 1, 1.5, 2),
           color = "grey50", linewidth = 0.3, linetype = "dashed", alpha = 0.6) +

# Highlight driest and wettest periods
geom_vline(xintercept = drought_stats$min_date, color = "#8B0000", linetype = "dotdash", linewidth = 

geom_vline(xintercept = drought_stats$max_date, color = "#00008B", linetype = "dotdash", linewidth = 

# KEY INSIGHTS annotation box
annotate("rect",
         xmin = as.Date("2004-05-01"), xmax = as.Date("2013-03-01"),
         ymin = 2.3, ymax = 3.2,
         fill = "white", color = "grey60", alpha = 0.9, linewidth = 0.5) +

annotate("text", x = as.Date("2008-01-01"), y = 3.0,
       label = paste0("KEY INSIGHTS (", min(year(rdi_36tbl$date)), "-", max(year(rdi_36tbl$date)), ")
       size = 5, fontface = "bold", hjust = 0.5) +


annotate("text", x = as.Date("2004-05-01"), y = 2.8,
         label = paste0("• Drought conditions: ", drought_stats$drought_pct, "% of period"),
         size = 5, hjust = 0, color = "#CC0000") +

annotate("text", x = as.Date("2004-05-01"), y = 2.6,
         label = paste0("• Severe drought: ", drought_stats$severe_drought_pct, "% of period"),
         size = 5, hjust = 0, color = "#8B0000", fontface = "bold") +

annotate("text", x = drought_stats$min_date, y = -2.9,
         label = paste0(format(drought_stats$min_date, "%Y %b"), " Event Extreme",
                        " (RDI: ", round(drought_stats$min_rdi, 1), ") •"),
         size = 5, hjust = 1, color = "grey30", fontface = "bold") +

annotate("text", x = drought_stats$max_date, y = drought_stats$max_rdi + 0.2,
         label = paste0("• Wettest event: ", format(drought_stats$max_date, "%b %Y"),
                        " (RDI: ", round(drought_stats$max_rdi, 1), ")"),
         size = 5, hjust = 0, color = "grey30", fontface = "bold") +

# Start and end of analysis period
annotate("text", x = min(rdi_36tbl$date), y = 3,
         label = "Start of analysis", hjust = 0, size = 5, color = "grey40", fontface = "bold") +
annotate("text", x = max(rdi_36tbl$date), y = 3,
         label = "End of analysis", hjust = 0.9, size = 5, color = "grey50", fontface = "bold") +

# Scales
scale_x_date(
  date_breaks = "5 years",
  date_labels = "%Y",
  limits = as.Date(c("1990-01-01", "2025-05-28")),
  expand = expansion(mult = c(0.01, 0.01))
) +

scale_y_continuous(
  limits = c(-3.5, 3.5),
```

```r
    breaks = seq(-3.5, 3, 0.5),
    expand = expansion(mult = c(0.01, 0.01))
  ) +

  scale_fill_manual(
    name = "Condition",
    values = drought_colors,
    guide = guide_legend(
      title.position = "top",
      title.hjust = 0.5,
      nrow = 1,
      reverse = TRUE
    )
  ) +

  # Labels and theme
  labs(
    title = "Drought Severity in Hamburg Fuhlsbüttel (1990-2025*)",
    subtitle = paste0("RDI (36-Month Scale) | ", stats_text),
    caption = "Source: DWD ",
    x = "",
    y = ""
  ) +

  theme_minimal(base_size = 12) +
  theme(
    panel.grid.minor.x = element_blank(),
    panel.grid.major.x = element_line(color = "grey90", linewidth = 0.5),
    panel.grid.minor.y = element_blank(),
    panel.grid.major.y = element_line(color = "grey90", linewidth = 0.5),
    plot.title = element_text(face = "bold", size = 22, margin = margin(t = 10, b = 15)),
    plot.subtitle = element_text(color = "grey60", size = 18, margin = margin(b = 10)),
    plot.caption = element_text(color = "grey60", size = 15, hjust = 1, margin = margin(t = 10)),
    axis.text.x = element_text(angle = 45, hjust = 1, size = 15, colour = "grey10", face = "bold"),
    axis.text.y = element_text(size = 15, colour = "gray10", face = "bold"),
    axis.title.y = element_text(size = 16, margin = margin(r = 10)),
    legend.position = "top",
    legend.title = element_text(size = 12, face = "bold"),
    legend.text = element_text(size = 9),
    legend.key.size = unit(1, "cm")
  )
```



Drought Severity in Hamburg Fuhlsbüttel (199
RDI (36−Month Scale) | Drought conditions: 37.2% | Severe
Extremely Wet
Very Wet
Moderately Wet
Slightly Wet
Start of analysis
. Wettest event: Jun 1995 (RDI: 2.7)
KEY INS
. Drought c
. Severe d
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
−3.0
−3.5
1994
1999
2004

The 36-month RDI shows that nearly two-fifths of the period was affected by long-term drought conditions. The most intense and prolonged drought occurred between 2014 and 2022, with the lowest RDI in January 2021 (-2.9) — a reflection of sustained precipitation deficits and increased evapotranspiration. In contrast, the early 1990s experienced the longest and strongest wet period, peaking in June 1995 (RDI = 2.7). These findings highlight a transition from predominantly wet to persistently dry long-term hydrological conditions over the last 30 years.

```r
library(Kendall)

# Ensure the data is in time order and remove NAs
rdi_36_values <- rdi_36tbl |> arrange(date) |> pull(RDI_36) |> na.omit()

# Run Mann-Kendall test
mk_test_36 <- MannKendall(rdi_36_values)
print(mk_test_36)
```

```
## tau = -0.438, 2-sided pvalue =< 2.22e-16
```

The Mann-Kendall trend analysis of the 36-month RDI series reveals a strong and statistically significant downward trend (= -0.438, $p < 0.0001$), indicating a persistent and intensifying shift toward long-term drought conditions in Hamburg Fuhlsbüttel over the period 1990–2025. This scale, which reflects hydrological droughts and multi-year water deficits, highlights a marked departure from the relatively wet conditions of the early 1990s. The trend corroborates observed extreme drought events, particularly the one peaking in January 2021 (RDI = -2.9), and suggests a long-term decline in regional water availability.

48 month RDI scale

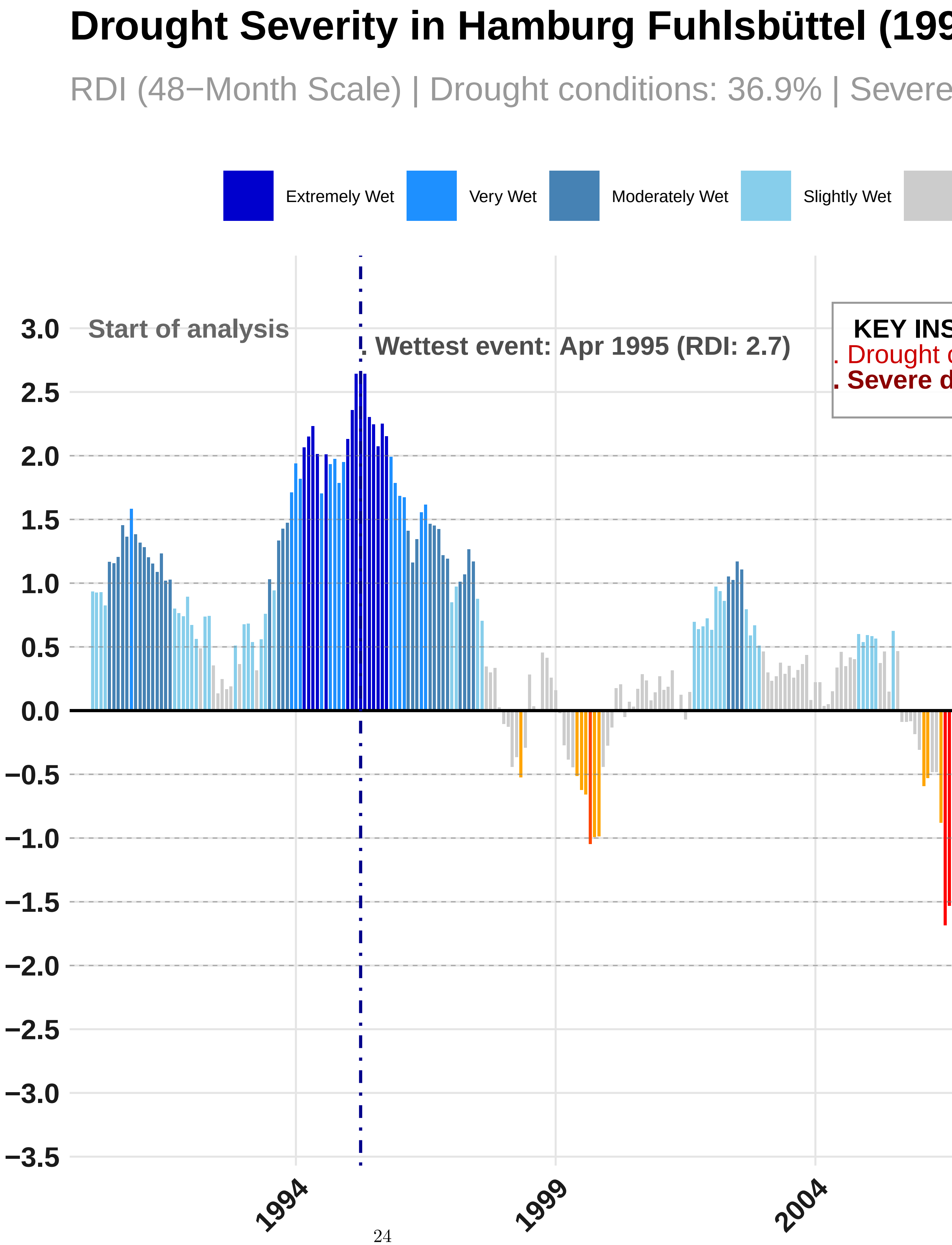
```r
# Data preparation
rdi_48tbl <- monthly_rdist_tbl |>
  dplyr::select(date, RDI_48) |>
  drop_na() |>
  filter(date > "1989-12-01") |>
  dplyr::mutate(
    # Original sign classification
    sign = ifelse(RDI_48 > 0, "pos", "neg"),

    drought_category = case_when(
      RDI_48 >= 2.0 ~ "Extremely Wet",
      RDI_48 >= 1.5 & RDI_48 < 2.0 ~ "Very Wet",
      RDI_48 >= 1.0 & RDI_48 < 1.5 ~ "Moderately Wet",
      RDI_48 >= 0.5 & RDI_48 < 1.0 ~ "Slightly Wet",
      RDI_48 >= -0.5 & RDI_48 < 0.5 ~ "Normal",
      RDI_48 >= -1.0 & RDI_48 < -0.5 ~ "Mild Drought",
      RDI_48 >= -1.5 & RDI_48 < -1.0 ~ "Moderate Drought",
      RDI_48 >= -2.0 & RDI_48 < -1.5 ~ "Severe Drought",
      TRUE ~ "Extreme Drought"  # RDI_48 < -2.0
    ),

    # Create ordered factor for proper legend ordering
    drought_category = factor(drought_category, levels = c(
```

```r
      "Extreme Drought", "Severe Drought", "Moderate Drought",
      "Mild Drought", "Normal", "Slightly Wet",
      "Moderately Wet", "Very Wet", "Extremely Wet"
    ))
  )

# Calculate key statistics for plot annotations
drought_stats <- rdi_48tbl |>
  summarise(
    total_months = n(),
    drought_months = sum(RDI_48 < -0.5),
    severe_drought_months = sum(RDI_48 < -1.5),
    wet_months = sum(RDI_48 > 0.5),
    min_rdi = min(RDI_48),
    max_rdi = max(RDI_48),
    min_date = date[which.min(RDI_48)],
    max_date = date[which.max(RDI_48)],
    drought_pct = round(drought_months / total_months * 100, 1),
    severe_drought_pct = round(severe_drought_months / total_months * 100, 1),
    wet_pct = round(wet_months / total_months * 100, 1)
  )

# Create statistical summary for subtitle
stats_text <- paste0(
  "Drought conditions: ", drought_stats$drought_pct, "% | ",
  "Severe drought: ", drought_stats$severe_drought_pct, "% | ",
  "Wettest: ", format(drought_stats$max_date, "%b %Y"), " (", round(drought_stats$max_rdi, 1), ") | ",
  "Driest: ", format(drought_stats$min_date, "%b %Y"), " (", round(drought_stats$min_rdi, 1), ")"
)

# Define color palette for drought categories
drought_colors <- c(
  "Extreme Drought" = "#8B0000",      # Dark Red
  "Severe Drought" = "#FF0000",       # Red
  "Moderate Drought" = "#FF4500",     # Orange Red
  "Mild Drought" = "#FFA500",         # Orange
  "Normal" = "#CCCCCC",               # Light Gray
  "Slightly Wet" = "#87CEEB",         # Sky Blue
  "Moderately Wet" = "#4682B4",       # Steel Blue
  "Very Wet" = "#1E90FF",             # Dodger Blue
  "Extremely Wet" = "#0000CD"         # Medium Blue
)
```

```r
# Create the plot
ggplot(rdi_48tbl, aes(x = date, y = RDI_48, fill = drought_category)) +
  geom_col(show.legend = TRUE, width = 20) +
  geom_hline(yintercept = 0, color = "black", linewidth = 0.8) +

  # Add reference dashed lines
  geom_hline(yintercept = c(-2, -1.5, -1, -0.5, 0.5, 1, 1.5, 2),
             color = "grey50", linewidth = 0.3, linetype = "dashed", alpha = 0.6) +

  # Highlight driest and wettest periods
```

```r
geom_vline(xintercept = drought_stats$min_date, color = "#8B0000", linetype = "dotdash", linewidth = 0

geom_vline(xintercept = drought_stats$max_date, color = "#00008B", linetype = "dotdash", linewidth = 0

# KEY INSIGHTS annotation box
annotate("rect",
         xmin = as.Date("2004-05-01"), xmax = as.Date("2013-03-01"),
         ymin = 2.3, ymax = 3.2,
         fill = "white", color = "grey60", alpha = 0.9, linewidth = 0.5) +

annotate("text", x = as.Date("2008-01-01"), y = 3.0,
        label = paste0("KEY INSIGHTS (", min(year(rdi_48tbl$date)), "-", max(year(rdi_48tbl$date)), ")
        size = 5, fontface = "bold", hjust = 0.5) +


annotate("text", x = as.Date("2004-05-01"), y = 2.8,
         label = paste0("• Drought conditions: ", drought_stats$drought_pct, "% of period"),
         size = 5, hjust = 0, color = "#CC0000") +

annotate("text", x = as.Date("2004-05-01"), y = 2.6,
         label = paste0("• Severe drought: ", drought_stats$severe_drought_pct, "% of period"),
         size = 5, hjust = 0, color = "#8B0000", fontface = "bold") +

annotate("text", x = drought_stats$min_date, y = -2.9,
         label = paste0(format(drought_stats$min_date, "%Y %b"), " Event Extreme",
                        " (RDI: ", round(drought_stats$min_rdi, 1), ") •"),
         size = 5, hjust = 1, color = "grey30", fontface = "bold") +

annotate("text", x = drought_stats$max_date, y = drought_stats$max_rdi + 0.2,
         label = paste0("• Wettest event: ", format(drought_stats$max_date, "%b %Y"),
                        " (RDI: ", round(drought_stats$max_rdi, 1), ")"),
         size = 5, hjust = 0, color = "grey30", fontface = "bold") +

# Start and end of analysis period
annotate("text", x = min(rdi_48tbl$date), y = 3,
         label = "Start of analysis", hjust = 0, size = 5, color = "grey40", fontface = "bold") +
annotate("text", x = max(rdi_48tbl$date), y = 3,
         label = "End of analysis", hjust = 0.9, size = 5, color = "grey50", fontface = "bold") +

# Scales
scale_x_date(
  date_breaks = "5 years",
  date_labels = "%Y",
  limits = as.Date(c("1990-01-01", "2025-05-28")),
  expand = expansion(mult = c(0.01, 0.01))
) +

scale_y_continuous(
  limits = c(-3.5, 3.5),
  breaks = seq(-3.5, 3, 0.5),
  expand = expansion(mult = c(0.01, 0.01))
) +
```

```r
scale_fill_manual(
  name = "Condition",
  values = drought_colors,
  guide = guide_legend(
    title.position = "top",
    title.hjust = 0.5,
    nrow = 1,
    reverse = TRUE
  )
) +

# Labels and theme
labs(
  title = "Drought Severity in Hamburg Fuhlsbüttel (1990-2025*)",
  subtitle = paste0("RDI (48-Month Scale) | ", stats_text),
  caption = "Source: DWD ",
  x = "",
  y = ""
) +

theme_minimal(base_size = 12) +
theme(
  panel.grid.minor.x = element_blank(),
  panel.grid.major.x = element_line(color = "grey90", linewidth = 0.5),
  panel.grid.minor.y = element_blank(),
  panel.grid.major.y = element_line(color = "grey90", linewidth = 0.5),
  plot.title = element_text(face = "bold", size = 22, margin = margin(t = 10, b = 15)),
  plot.subtitle = element_text(color = "grey60", size = 18, margin = margin(b = 10)),
  plot.caption = element_text(color = "grey60", size = 15, hjust = 1, margin = margin(t = 10)),
  axis.text.x = element_text(angle = 45, hjust = 1, size = 15, colour = "grey10", face = "bold"),
  axis.text.y = element_text(size = 15, colour = "gray10", face = "bold"),
  axis.title.y = element_text(size = 16, margin = margin(r = 10)),
  legend.position = "top",
  legend.title = element_text(size = 12, face = "bold"),
  legend.text = element_text(size = 9),
  legend.key.size = unit(1, "cm")
)
```



Drought Severity in Hamburg Fuhlsbüttel (199
RDI (48−Month Scale) | Drought conditions: 36.9% | Severe
Extremely Wet
Very Wet
Moderately Wet
Slightly Wet
Start of analysis
Wettest event: Apr 1995 (RDI: 2.7)
KEY INS
. Drought c
. Severe d
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
−3.0
−3.5
1994
1999
2004

```r
rdi_48_values <- rdi_48tbl |> arrange(date) |> pull(RDI_48) |> na.omit()

# Run Mann-Kendall test
mk_test_48 <- MannKendall(rdi_48_values)
print(mk_test_48)
```

```
## tau = -0.563, 2-sided pvalue =< 2.22e-16
```

The 48-month RDI shows a persistent and intensifying long-term drought trend in Hamburg Fuhlsbüttel from 1990 to 2025. Nearly 37% of the period exhibits negative RDI values below -0.5, and 8.2% qualifies as severe drought. The most extreme drought occurred in January 2022 (RDI = -2.7), following a decade of steadily declining multi-year moisture conditions. The Mann-Kendall test confirms a strong and statistically significant drying trend (= -0.563, p < 0.0001), signaling major long-term hydrological stress in the region. In contrast, the early 1990s, particularly April 1995, marked a period of sustained wet conditions (RDI = 2.7), a level that has not been approached since.

```r
rdi_summary_tbl <- tibble::tibble(
  `RDI Scale` = c("12-Month", "24-Month", "36-Month", "48-Month"),
  `Drought (%)` = c("28.2%", "37.6%", "37.2%", "36.9%"),
  `Severe Drought (%)` = c("7.8%", "4.7%", "6.1%", "8.2%"),
  `Wettest Event` = c("May 1994 (2.6)", "May 1995 (3.0)", "Jun 1995 (2.7)", "Apr 1995 (2.7)"),
  `Driest Event` = c("Jan 2019 (-3.0)", "Jan 2020 (-2.4)", "Jan 2021 (-2.9)", "Jan 2022 (-2.7)"),
  `Kendall's Tau` = c("-0.163", "-0.282", "-0.438", "-0.563"),
  `Trend Significance` = c(" ", " ", " ", "  ")
)

print(rdi_summary_tbl)
```

RDI Summary

```
## # A tibble: 4 x 7
##   `RDI Scale` `Drought (%)` `Severe Drought (%)` `Wettest Event` `Driest Event`
##   <chr>       <chr>         <chr>                <chr>           <chr>
## 1 12-Month    28.2%         7.8%                 May 1994 (2.6)  Jan 2019 (-3.0)
## 2 24-Month    37.6%         4.7%                 May 1995 (3.0)  Jan 2020 (-2.4)
## 3 36-Month    37.2%         6.1%                 Jun 1995 (2.7)  Jan 2021 (-2.9)
## 4 48-Month    36.9%         8.2%                 Apr 1995 (2.7)  Jan 2022 (-2.7)
## # i 2 more variables: `Kendall's Tau` <chr>, `Trend Significance` <chr>
```

```r
knitr::kable(rdi_summary_tbl, caption = "Comparative Summary of RDI Indices (1990-2025)")
```

Table 1: Comparative Summary of RDI Indices (1990–2025)

RDI Scale	Drought (%)	Severe Drought (%)	Wettest Event	Driest Event	Kendall's Tau	Trend Significance
12-Month	28.2%	7.8%	May 1994 (2.6)	Jan 2019 (-3.0)	-0.163	

Comparative Summary of RDI Indices
Hamburg Fuhlsbüttel (1990–2025)

RDI Scale	Drought (%)	Severe Drought (%)	Wettest Event	Driest Event	Kendall's Tau	Trend Significance
12-Month	28.2%	7.8%	May 1994 (2.6)	Jan 2019 (-3.0)	-0.163	
24-Month	37.6%	4.7%	May 1995 (3.0)	Jan 2020 (-2.4)	-0.282	
36-Month	37.2%	6.1%	Jun 1995 (2.7)	Jan 2021 (-2.9)	-0.438	
48-Month	36.9%	8.2%	Apr 1995 (2.7)	Jan 2022 (-2.7)	-0.563	

RDI Scale	Drought (%)	Severe Drought (%)	Wettest Event	Driest Event	Kendall's Tau	Trend Significance
24-Month	37.6%	4.7%	May 1995 (3.0)	Jan 2020 (-2.4)	-0.282	
36-Month	37.2%	6.1%	Jun 1995 (2.7)	Jan 2021 (-2.9)	-0.438	
48-Month	36.9%	8.2%	Apr 1995 (2.7)	Jan 2022 (-2.7)	-0.563	

```r
library(gt)

rdi_summary_tbl |>
  gt() |>
  tab_header(
    title = "Comparative Summary of RDI Indices",
    subtitle = "Hamburg Fuhlsbüttel (1990-2025)"
  ) |>
  tab_options(
    table.font.size = 12,
    heading.title.font.size = 16,
    heading.subtitle.font.size = 14
  )
```

```r
library(flextable)

rdi_summary_tbl |>
  flextable() |>
  autofit() |>
  set_caption("Comparative Summary of RDI Indices - Hamburg Fuhlsbüttel (1990-2025)")
```

Table 2: Comparative Summary of RDI Indices – Hamburg Fuhlsbüttel (1990–2025)

RDI Scale	Drought (%)	Severe Drought (%)	Wettest Event	Driest Event	Kendall's Tau
12-Month	28.2%	7.8%	May 1994 (2.6)	Jan 2019 (-3.0)	-0.163
24-Month	37.6%	4.7%	May 1995 (3.0)	Jan 2020 (-2.4)	-0.282
36-Month	37.2%	6.1%	Jun 1995 (2.7)	Jan 2021 (-2.9)	-0.438
48-Month	36.9%	8.2%	Apr 1995 (2.7)	Jan 2022 (-2.7)	-0.563